

CY 302: Principles of Organic Chemistry

Course Code	CY 302
Course Name	Principles of Organic Chemistry
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BS Chemical Sciences
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Bonding & Aromaticity:** Atomic orbitals and their overlaps, bonding of different types – s-bond, p-bond, Bond dissociation energy, Bond order and multiplicity, Hybridization, VSEPR theory, Bent's rule, Dipole moment, Molecular orbital (MO) theory, Electronic effects: inductive & field effects, Mesomeric effects, Hyperconjugation, Resonance. Aromaticity, antiaromaticity, and homoaromaticity; Hückel's rule, aromatic ring currents; examples of nonbenzenoid aromatic and antiaromatic compounds. (10 Hours)
- **Reaction Kinetics and Reactive Intermediates:** Energetics of a chemical reaction, Transition state, Hammond's postulate, Hammett equation, Arrhenius equation, Effect of a catalyst, Kinetic Isotope Effect (primary and secondary), Isotope scrambling, Structure, stability and reactivity of carbocations, carbanions, free radicals, carbenes, and nitrenes. (8 Hours)
- **Acid-Base Theory:** Acidity, basicity, and pK_a, Brønsted & Lowry concept, Lewis concept, The definition of pK_a, Basicity, Factors that influence the acidity and basicity, HSAB Principle, Keto-enol tautomerism. (4 Hours)
- **Principles of Stereochemistry:** Baeyer's strain theory, Pitzer strain (torsional strain) and conformational analysis (up to cyclohexane), geometrical isomerism (E/Z), optical isomerism, projections, CIP rules (R/S nomenclature of acyclic and cyclic molecules); nomenclature – threo and erythro, syn and anti, endo and exo, and meso and d/l; Introduction to chirality and its origin. (10 Hours)
- **Organic Reactions:** Carbon-carbon bond forming reactions, Olefination reactions, Reduction & oxidation reactions, aromatic substitution reactions (electrophilic, nucleophilic, etc.), chemistry of carbonyl compounds, alkenes and alkynes. (10 Hours)

Text Books

1. Clayden, J., Greeves, N., Warren, S., Wothers, S., **Organic Chemistry**, Oxford University Press, 2001.
2. Eliel, E. L., Wilen, S. H., Doyle, M. P., **Basic Organic Stereochemistry**, John Wiley and Sons, 2001.
3. Smith, M. B. and March, J., **Advanced Organic Chemistry**, Wiley Interscience, 2007.
4. D. Nasipuri, **Stereochemistry of Organic Compounds-Principle and Applications**, 4 Revised Edition, New Academic Science, 2012.
5. P. Sykes, **A Guidebook to Mechanism in Organic Chemistry**, 7th Edition, Addison- Wesley, 2003.

References

1. H. O. House, **Modern Synthetic Reactions**, W.A. Benjamin Inc., 1972.
2. A. Jacobs, **Understanding Organic Reaction Mechanism**, Cambridge 1998.